

Arhan Chaudhary — arhan.sh

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EDUCATION

Purdue University (B.S. Computer Science, Minor Electrical Computer Engineering)

Class of 2027

- Relevant Coursework: Compilers, Data Structures and Algorithms, Computer Architecture, Linear Algebra, Electrical Engineering Fundamentals I & II • GPA: 3.7

SupportVectors AI Bootcamp

Summer 2027

- Other Coursework: Foundations of AI (taken at supportvectors.ai)

EXPERIENCE

EnCharge AI (enchargeai.com), Compiler Engineer Intern — Hybrid (Santa Clara)

May 2025 - August 2025

- Developed an NPU model binary disassembler in C++ for custom chip hardware, used for analysis in the compiler.
- Designed an efficient and queryable binary interface tool for visualizing hardware simulation output, emphasizing accounting for feedback from prospective internal users. Integrated into the hardware simulation workflow.

SPAN (span.io), Backend Software Engineer Intern — On-site (San Francisco)

May 2024 - August 2024

- Developed an internal tool for aggregating data into a database sink as final project, with extensive documentation on user usage.
- Implemented a highly customizable retry handler for an internal workflow orchestration framework. Integrated into production.
- Troubleshooted issues with DataDog and performed general sysadmin debugging.

The Data Mine (datamine.purdue.edu), Data Science Researcher — On-site (Purdue University)

Jan 2024 - April 2024

- Developed a model for Cisco to analyze and predict product data, utilizing MLE and Bayesian statistics with a team of eight.
- Presented at Corporate Partners Symposium as part of Purdue University's Data Mine Corporate Partners program.

PROJECTS

Qter — github.com/qter-project and qter.dev

September 2024 - Present

- Human-friendly and turing-complete computer system that uses the Rubik's Cube to physically perform computations.
- Architected custom programming language and compiler, featuring support for macros, basic math, and call stacks.
- Developed a suite of many efficient puzzle-agnostic tools in Rust, including a domain specific optimal twisty puzzle solver, which utilizes SIMD programming, multithreading, and novel mathematical observations.
- Built a custom Rubik's Cube robot to execute Qter programs with three project members. Published 50+ page academic preprint.

NAND — github.com/ArhanChaudhary/NAND and nand.arhan.sh

June 2023 - July 2024

- Programmable 16-bit computer made entirely from NAND gates emulated on the web. 575+ GitHub stars, #1 Hacker News.
- Features its own IDE, programming language, operating system, kernel, compiler, and CPU. Created 6.5k word writeup.
- Compiled from Rust to WebAssembly on multithreaded web workers and served on a Svelte frontend.

TimeWeb — github.com/ArhanChaudhary/TimeWeb

Oct 2019 - June 2023

- Conceptualized a passion project and developed a homework management web app that visualizes, quantifies, and prioritizes daily school assignments using Python/Django and JavaScript. Wrote 20k+ lines of code over 3.5 years.
- Utilized Google Classroom API and OAuth2, created user guide, managed database and migrations, and organized 35+ releases.
- Launched to high school in 2022 with multiple class presentations to 250+ students, gaining 700+ users.

ACHIEVEMENTS

- Open source: 6k+ public GitHub commits. Submitted 30+ pull requests and 30+ issues to various open source repos.
- CTF competitions: 1st place Starphish CTF at DEFCON 31 • 1st place b01lersctf 2023 Purdue division • 10th place SquareCTF 2023 • Top 50 national CTftime team 2023.
- Personal technical blogging: 15k+ words, reached #3 on Lobste.rs, featured on This Week in Rust (arhan.sh/blog)
- Hackathon: 3rd place "HelloWorld @ Purdue Fall 2023" (devpost.com/software/gitme-amuryi)
- Rubik's Cube: Personal best solve time of 6.48s, can also solve blindfolded.

TECHNICAL SKILLS

Languages Rust, C, C++, JavaScript, TypeScript, Python, Java • **Cloud** Cloudflare, AWS, GCP • **Frameworks** Svelte, React, Flask, Astro • **Others** Nix, Snowflake, Terraform, Protobuf, PostgreSQL, Git, Technical writing